Data analysis - Cadaver Measurements

Libraries and Functions

```r
# Clear envirnonment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

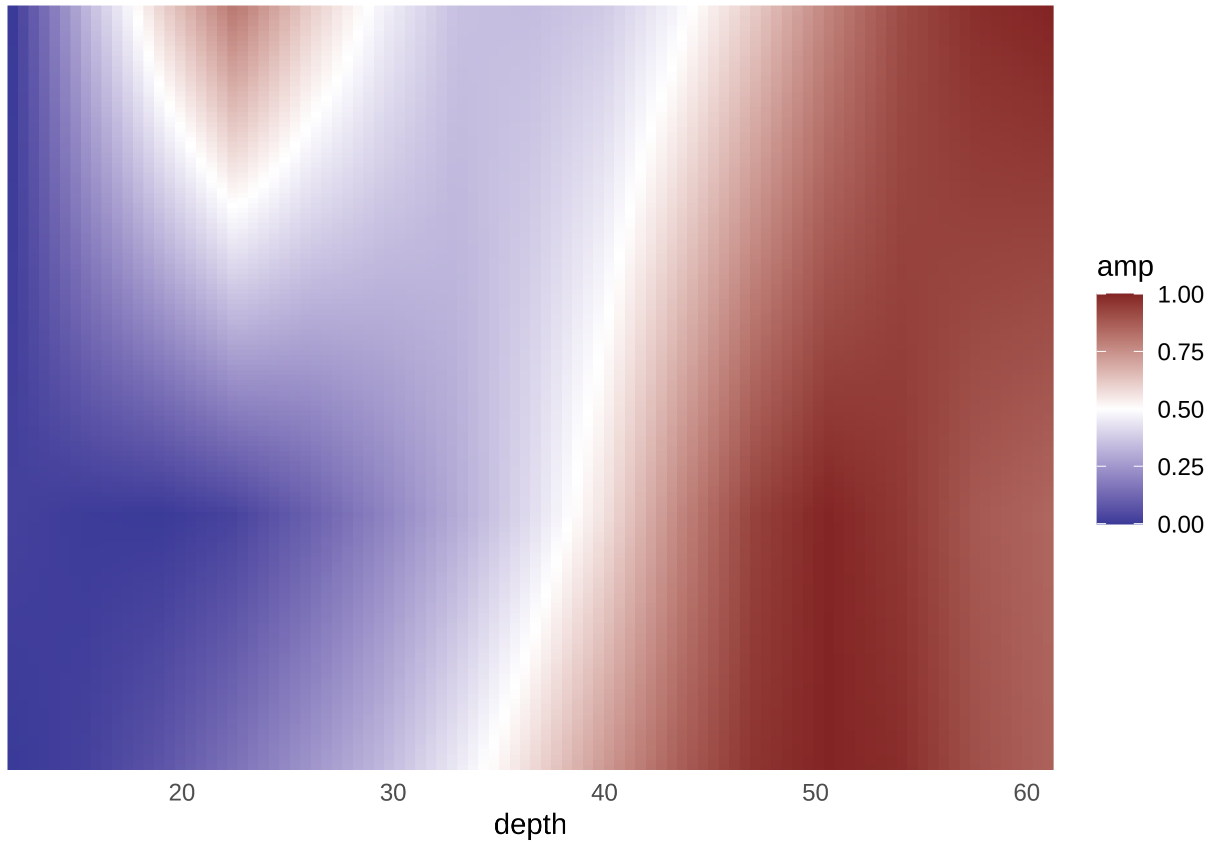
Import data and Organize it

```r
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Cadaver")
data <- read.csv("ElectrodeData.csv")

# add cortical (contacts 15 - 9) and hipp (contacts 8 - 1) regions
data$contact <- as.numeric(data$contact)
data$region[data$contact>8] <- "Cortex"
data$region[data$contact<9] <- "Hipp"

# Assign factors

data$StimType <- factor(data$StimType)
data$elect <- factor(data$elect)
data$type <- factor(data$type)
data$region <- factor(data$region)


data <- data %>% rename(depth = distance)


Plane_elec <- data

Plane_elec$distance[Plane_elec$elect=="a"] <- 0.5  ## anterior electrode
Plane_elec$distance[Plane_elec$elect=="b"] <- 1.5  # mid electrode
Plane_elec$distance[Plane_elec$elect=="c"] <- 2 # posterior electrode
```

Plot data - Figure S2

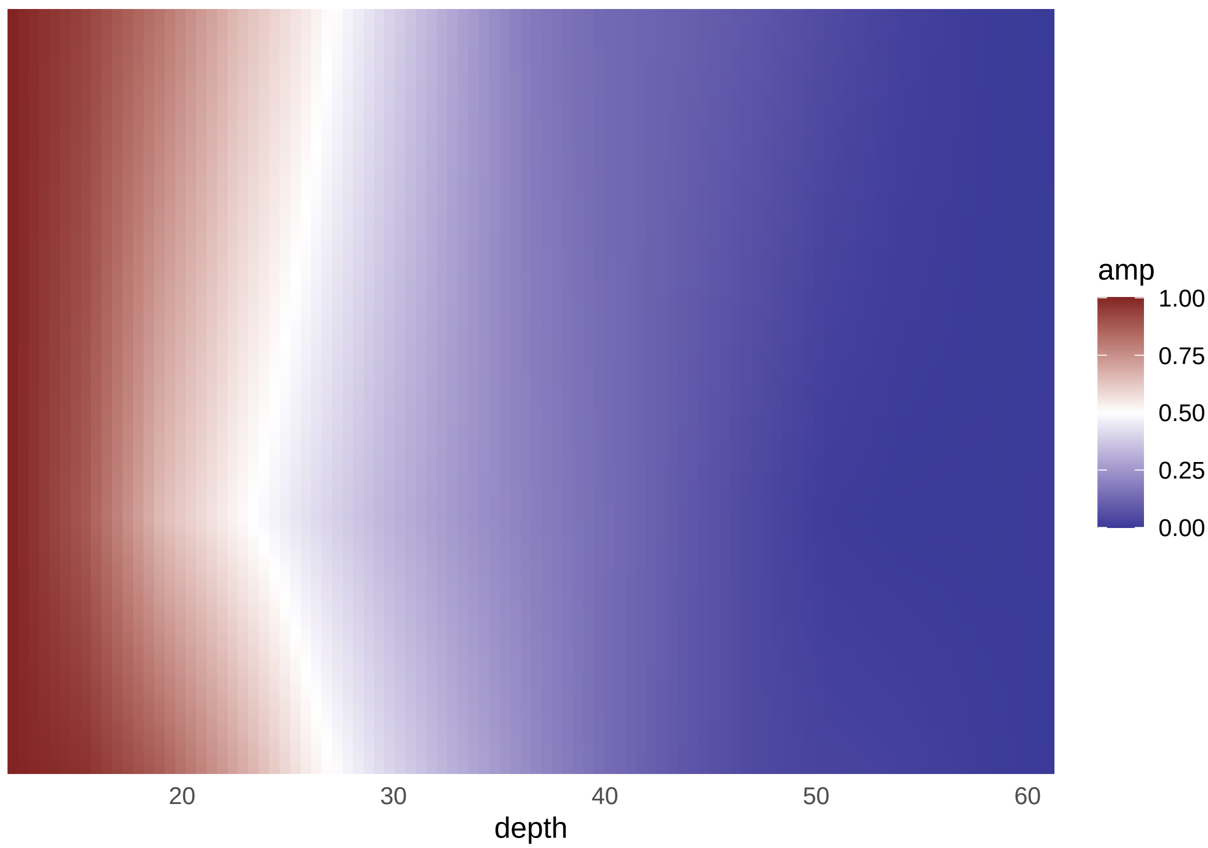
```r
## Map plots
TI <- subset(Plane_elec, StimType=="TI")
```

```r
# ---- TI 1:1 - Figure1Gi
amp_interp_depth <- tidyr::crossing(
  tibble(distance = unique(TI$distance)),
  tibble(depth = seq(12, 61, length.out = 100))
) %>%
  group_by(distance) %>%
  mutate(mv_env_amp_norm = estimate_amp_by_distance(TI,distance[1], depth))


amp_raster <- tidyr::crossing(
  tibble(distance = seq(0.5,2, by = .01)),
  tibble(depth = unique(amp_interp_depth$depth))
) %>%
  group_by(depth) %>%
  mutate(amp = estimate_amp_by_depth(depth[1], distance))


p.amp_TI <- ggplot(amp_raster, aes(depth, distance, fill = amp)) +
  geom_raster() +
  scale_y_reverse() +
  scale_fill_gradient2(
    midpoint = 0.5,
    high = scales::muted("red"),
    low = scales::muted("blue")
  ) +
  coord_cartesian(expand = FALSE) +theme(
  axis.text.y = element_blank(),
  axis.ticks = element_blank()) + easy_remove_y_axis()

p.amp_TI
```



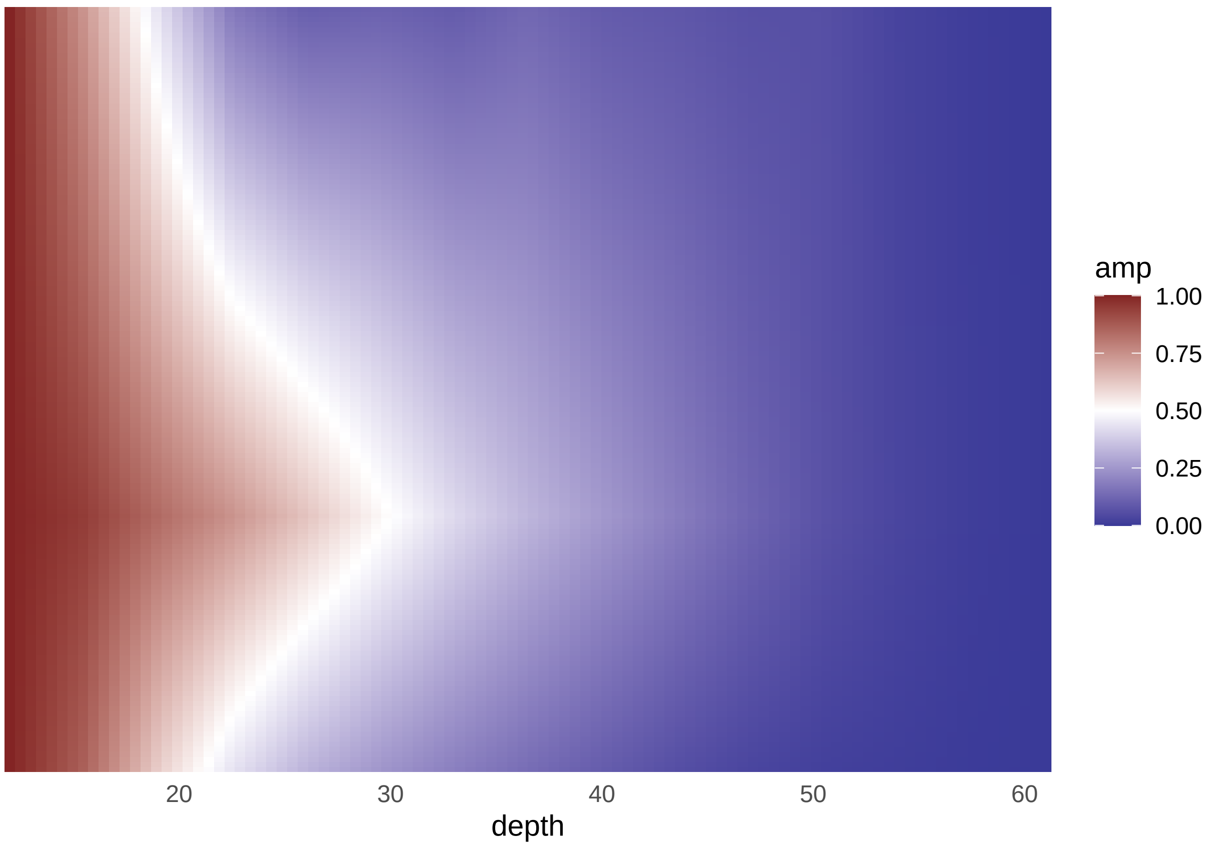

```
ggsave(p.amp_TI ,file="Figure_S2i.eps", width = 10, height = 5, dpi = 300, units = "cm", device = "eps")

####

# ---- HF 1:1 - Figure1Gii
amp_interp_depth <- tidyr::crossing(
  tibble(distance = unique(TI$distance)),
  tibble(depth = seq(12, 61, length.out = 100))
) %>%
  group_by(distance) %>%
  mutate(mv_env_max_norm = HF_estimate_amp_by_distance(TI,distance[1], depth))


amp_raster <- tidyr::crossing(
  tibble(distance = seq(0.5,2, by = .01)),
  tibble(depth = unique(amp_interp_depth$depth))
) %>%
  group_by(depth) %>%
  mutate(amp = HF_estimate_amp_by_depth(depth[1], distance))


p.HF_amp <- ggplot(amp_raster, aes(depth, distance, fill = amp)) +
  geom_raster() +
  scale_y_reverse() +
  scale_fill_gradient2(
    midpoint = 0.5,
```

```r
    high = scales::muted("red"),
    low = scales::muted("blue")
  ) +
  coord_cartesian(expand = FALSE) +theme(
  axis.text.y = element_blank(),
  axis.ticks = element_blank()) + easy_remove_y_axis()

p.HF_amp
```




```r
ggsave(p.HF_amp ,file="Figure_S2ii.eps", width = 10, height = 5, dpi = 300, units = "cm", device = "eps


# ---- LF (tACS 5 Hz) - Figure S1iii
LF <-  subset(Plane_elec, StimType=="LF")
amp_interp_depth <- tidyr::crossing(
  tibble(distance = unique(LF$distance)),
  tibble(depth = seq(12, 61, length.out = 100))
) %>%
  group_by(distance) %>%
  mutate(mv_env_max_norm = HF_estimate_amp_by_distance(LF,distance[1], depth))


amp_raster <- tidyr::crossing(
  tibble(distance = seq(0.5,2, by = .01)),
  tibble(depth = unique(amp_interp_depth$depth))
```

```
) %>%
  group_by(depth) %>%
  mutate(amp = HF_estimate_amp_by_depth(depth[1], distance))


p.LF_amp <- ggplot(amp_raster, aes(depth, distance, fill = amp)) +
  geom_raster() +
  scale_y_reverse() +
  scale_fill_gradient2(
    midpoint = 0.5,
    high = scales::muted("red"),
    low = scales::muted("blue")
  ) +
  coord_cartesian(expand = FALSE) +theme(
  axis.text.y = element_blank(),
  axis.ticks = element_blank()) + easy_remove_y_axis()

p.LF_amp
```

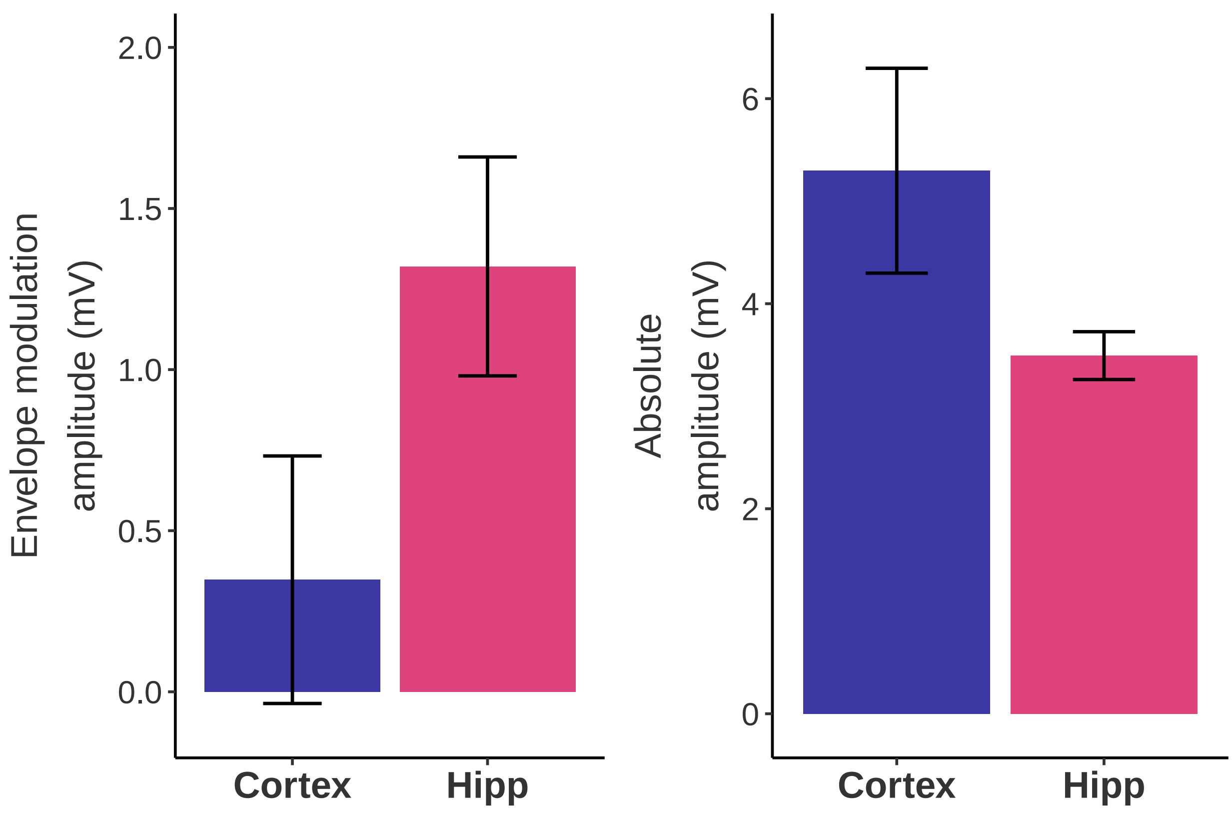



```
ggsave(p.LF_amp ,file="Figure_S2iii.eps", width = 10, height = 5, dpi = 300, units = "cm", device = "eps
```

Plot data - Figure 1G

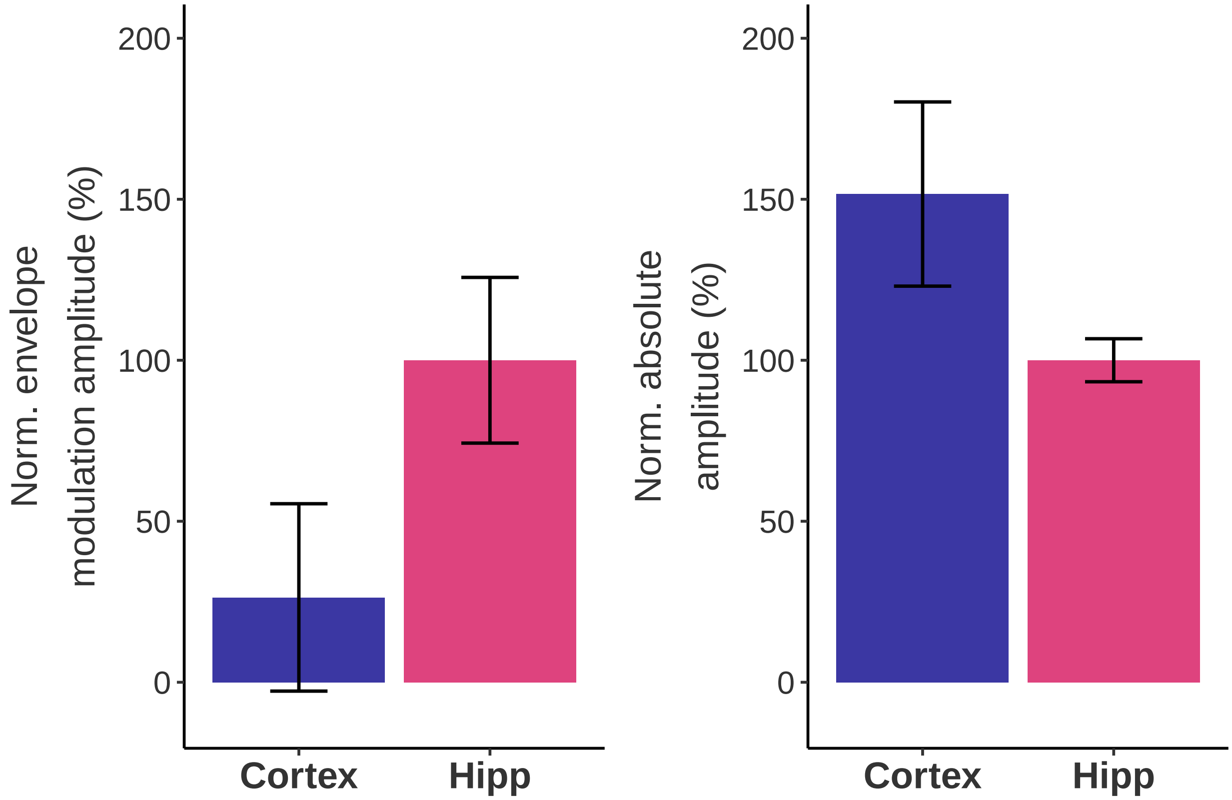
```
Plane_elec_TI <-subset(Plane_elec, StimType=="TI")


g.Plane_elec_region <- ddply(Plane_elec_TI, c("region","elect"), summarise,
                   HF_mean = mean(env_max),
                   HF_median = median(env_max),
                   HF_sd = sd(env_max),
                   HF_se = se(env_max),
                   TI_mean = mean(env_amp),
                   TI_median = median(env_amp),
                   TI_sd = sd(env_amp),
                   TI_se = se(env_amp),
                   d1_TI = mean(d1_env_amp),
                   d1_HF = mean(d1_env_max)
                   )


# Averages for plotting
g.Plane_elec <- ddply(Plane_elec_TI, c("region"), summarise,
                   HF_mean = mean(env_max),
                   HF_median = median(env_max),
                   HF_sd = sd(env_max),
                   HF_se = se(env_max),
                   TI_mean = mean(env_amp),
                   TI_median = median(env_amp),
                   TI_sd = sd(env_amp),
                   TI_se = se(env_amp),
                   d1_TI = mean(d1_env_amp),
                   d1_HF = mean(d1_env_max)
                   )

p.TI.elect_region <- ggplot(g.Plane_elec, aes(x = region, y = TI_median*10^-3)) +
       geom_bar(stat="identity", fill=c("#3b37a3", "#de437e")) +
  geom_errorbar(aes(ymin=(TI_median-TI_sd)*10^-3, ymax=(TI_median+TI_sd)*10^-3), width=0.3, size=.6,
                 position=position_dodge(.9)) +
        scale_y_continuous(
                          limits=c(-0.1, 2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x =  element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Env
```

```
p.HF.elect_region <- ggplot(g.Plane_elec, aes(x = region, y = HF_median*10^-3)) +
       geom_bar(stat="identity", fill=c("#3b37a3", "#de437e")) +
  geom_errorbar(aes(ymin=(HF_median-HF_sd)*10^-3, ymax=(HF_median+HF_sd)*10^-3), width=0.3, size=.6,
                position=position_dodge(.9)) +
        scale_y_continuous(
                          limits=c(-0.1, 6.5)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x =  element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Abs

ROI <- grid.arrange(p.TI.elect_region, p.HF.elect_region, nrow=1, ncol=2)
```



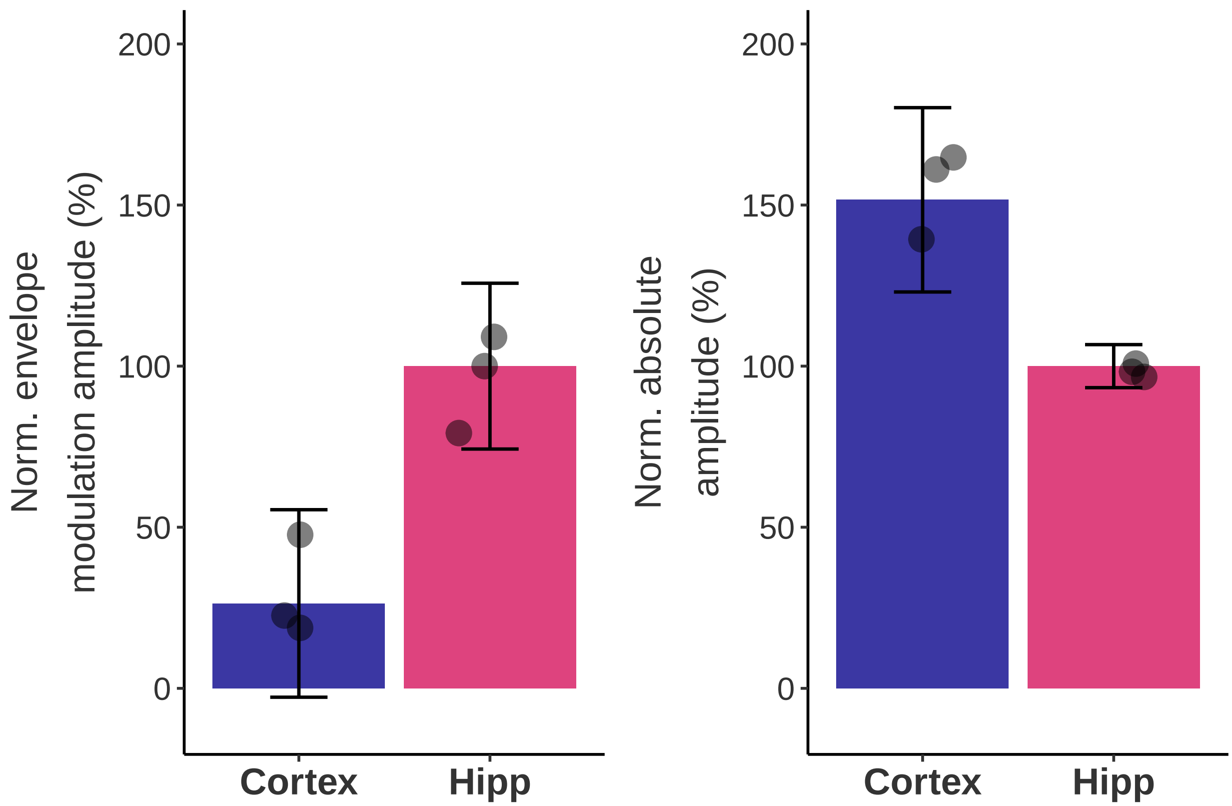
```r
### Normalise to hippocampus

norm <- ddply(g.Plane_elec, c("region"), summarise,
                     TI_norm = (TI_median)/(g.Plane_elec$TI_median[g.Plane_elec$region=="Hipp"])*100,
              TI_SDnorm = (TI_sd)/(g.Plane_elec$TI_median[g.Plane_elec$region=="Hipp"])*100,
              HF_norm = (HF_median)/(g.Plane_elec$HF_median[g.Plane_elec$region=="Hipp"])*100,
              HF_SDnorm = (HF_sd)/(g.Plane_elec$HF_median[g.Plane_elec$region=="Hipp"])*100
              )


p.TI.elect_region_norm <- ggplot(norm, aes(x = region, y = TI_norm)) +
       geom_bar(stat="identity", fill=c("#3b37a3", "#de437e")) +
  geom_errorbar(aes(ymin=(TI_norm-TI_SDnorm), ymax=(TI_norm+TI_SDnorm)), width=0.3, size=.6,
                 position=position_dodge(.9)) +
        scale_y_continuous(
                           limits=c(-10, 200)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x =  element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Nor



p.HF.elect_region_norm <- ggplot(norm, aes(x = region, y = HF_norm)) +
       geom_bar(stat="identity", fill=c("#3b37a3", "#de437e")) +
  geom_errorbar(aes(ymin=(HF_norm-HF_SDnorm), ymax=(HF_norm+HF_SDnorm)), width=0.3, size=.6,
                 position=position_dodge(.9)) +
        scale_y_continuous(
                           limits=c(-10, 200)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x =  element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Nor
```

```
ROI_norm <- grid.arrange(p.TI.elect_region_norm, p.HF.elect_region_norm, nrow=1, ncol=2)
```




```
#ggsave(ROI_norm, file="Figure_1H.tiff", width = 12, height = 8, dpi = 300, units = "cm", device = "tif


# ----------- Plots with electrode info -- each time the dots will be generated in a different location

g.norm.Plane_elec_region <- g.Plane_elec_region %>%
   mutate(TI_mean_norm = (TI_mean /g.Plane_elec$TI_mean[g.Plane_elec$region=="Hipp"])*100,
          TI_median_norm = (TI_median / g.Plane_elec$TI_median[g.Plane_elec$region=="Hipp"])*100,
          TI_sd_norm = (TI_sd / g.Plane_elec$TI_median[g.Plane_elec$region=="Hipp"])*100,
          HF_mean_norm = (HF_mean /g.Plane_elec$HF_mean[g.Plane_elec$region=="Hipp"])*100,
          HF_median_norm = (HF_median / g.Plane_elec$HF_median[g.Plane_elec$region=="Hipp"])*100,
          HF_sd_norm = (HF_sd / g.Plane_elec$HF_median[g.Plane_elec$region=="Hipp"])*100) %>%
   ungroup


p.TI.elect_region_norm2 <- ggplot()+
  geom_bar(data =norm, aes(x = region, y = TI_norm), stat="identity", fill=c("#3b37a3", "#de437e")) +
  geom_errorbar(data =norm, aes(x = region, ymin=(TI_norm-TI_SDnorm), ymax=(TI_norm+TI_SDnorm)), width=(
               position=position_dodge(.9)) +
  geom_point(data = g.norm.Plane_elec_region, aes(x=region, y=TI_median_norm), size=4, alpha=0.5, positi

      scale_y_continuous(
                      limits=c(-10, 200)) +
```

```
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x =  element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Norm


p.HF.elect_region_norm2 <- ggplot(norm, aes(x = region, y = HF_norm)) +
       geom_bar(stat="identity", fill=c("#3b37a3", "#de437e")) +
  geom_errorbar(aes(ymin=(HF_norm-HF_SDnorm), ymax=(HF_norm+HF_SDnorm)), width=0.3, size=.6,
                 position=position_dodge(.9)) +
   geom_point(data = g.norm.Plane_elec_region, aes(x=region, y=HF_median_norm), size=4, alpha=0.5, posi

        scale_y_continuous(
                          limits=c(-10, 200)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x =  element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Norm

ROI_norm <- grid.arrange(p.TI.elect_region_norm2, p.HF.elect_region_norm2, nrow=1, ncol=2)
```




```r
ggsave(ROI_norm, file="Figure_1g.eps", width = 12, height = 8, dpi = 300, units = "cm", device = "eps")

# -------------------------------------------------------------------------------
# Compare regions-- Table S1

g.Plane_elec_region$elect <- as.factor(g.Plane_elec_region$elect)


TI.m <- lmer(TI_median ~ region  + (1|elect), data = g.Plane_elec_region)
Anova(TI.m, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: TI_median
##             F Df Df.res  Pr(>F)
## region 27.533  1      2 0.03445 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
emmeans(TI.m, specs = pairwise ~ region, type = "response" )
```

```
## $emmeans
##  region emmean  SE df lower.CL upper.CL
##  Cortex    392 118  4     63.8      720
##  Hipp     1269 118  4    940.7     1597
```

```
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##  contrast      estimate  SE df t.ratio p.value
##  Cortex - Hipp     -877 167  2  -5.247  0.0345
##
## Degrees-of-freedom method: kenward-roger
```

```
HF.m <- lmer(HF_median ~ region +  (1|elect), data = g.Plane_elec_region, REML = TRUE)
Anova(HF.m, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: HF_median
##             F Df Df.res  Pr(>F)
## region 49.723  1      2 0.01952 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

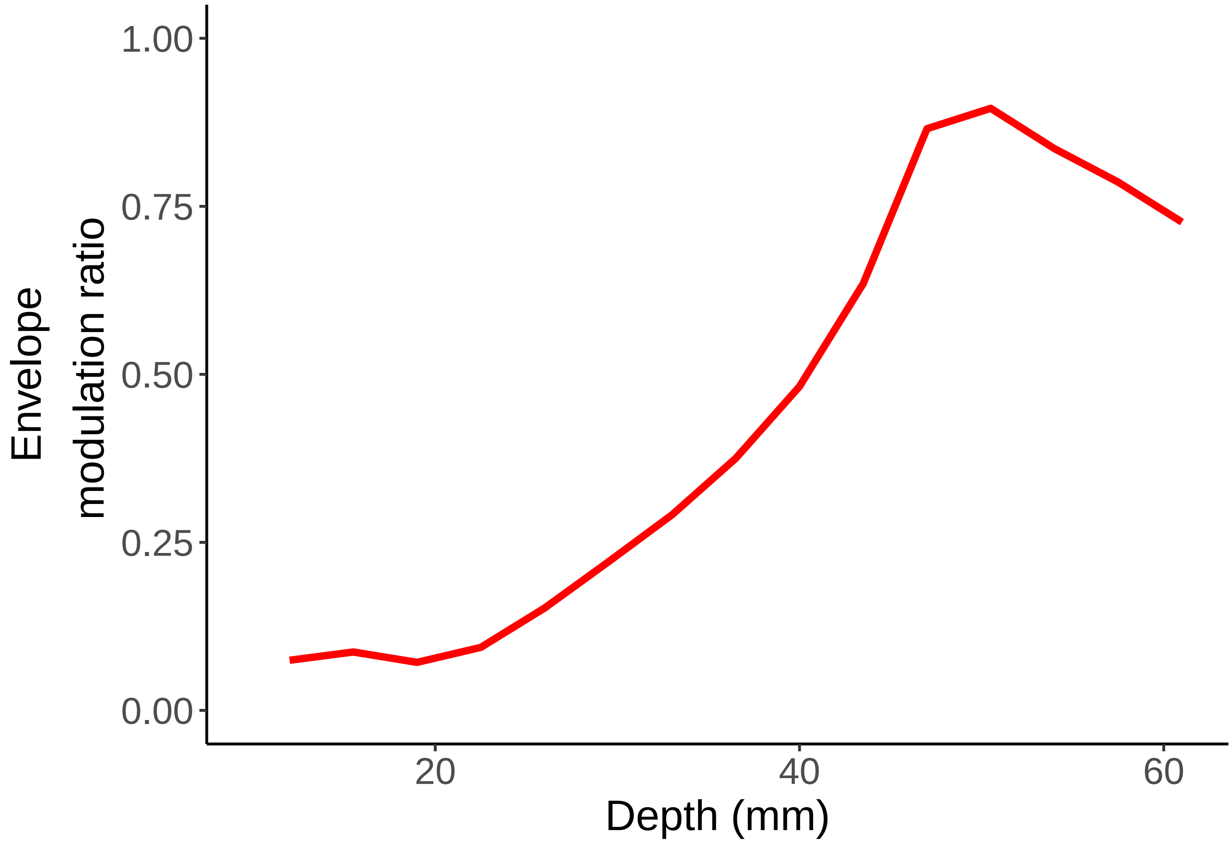
```
emmeans(HF.m, specs = pairwise ~ region, type = "response" )
```

```
## $emmeans
##  region emmean  SE df lower.CL upper.CL
##  Cortex   5417 198  4     4868     5967
##  Hipp     3444 198  4     2894     3993
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##  contrast      estimate  SE df t.ratio p.value
##  Cortex - Hipp     1974 280  2   7.051  0.0195
##
## Degrees-of-freedom method: kenward-roger
```

Plot data - Figure 1h

```
## Mod ratio plot - electrode b
Mod_ratio <- subset(Plane_elec, StimType=="TI" & elect=="b")

h_line_max <- round(max(Mod_ratio$mod_ratio),2)
h_line_min <- round(min(Mod_ratio$mod_ratio),2)

p.Mod_ratio <- ggplot(Mod_ratio, aes(x=depth, y=mod_ratio)) + geom_line(color = "red", size=1.3) +
  theme_bw() +
  theme(
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    axis.title.x = element_text(size = 16),
```

```
    axis.title.y = element_text(size = 16),
    panel.border = element_blank(), panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  scale_x_continuous(name="Depth (mm)", limits=c(10, 61)) +
  scale_y_continuous(limits=c(0, 1)) +
  theme(legend.position="none") + ylab(expression(atop("Envelope", paste(" modulation ratio"))))

p.Mod_ratio
```




```
ggsave(p.Mod_ratio,file="Figure_1h.eps", width = 7, height = 7, dpi = 300, units = "cm", device = "eps")
```

Electric Field

```
Elec_TI <-subset(Plane_elec, StimType=="TI" & elect=="b")
MaxElectricField <- max(Elec_TI$d1_env_amp)
DepthMax <- Elec_TI[which.max(Elec_TI[,"d1_env_amp"]),"depth"]


MaxElectricField_HF <- max(abs(Elec_TI$d1_env_max))
DepthMaxHF <- Elec_TI[which.min(Elec_TI[,"d1_env_max"]),"depth"]


p.ElecField <- ggplot(Elec_TI, aes(x=depth, y=d1_env_amp)) +
```

```
  geom_line(color = "red") +
  geom_line(data=Elec_TI, aes(x=depth, y=abs(d1_env_max)), color = "black")
```